

	DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Examination – Summer 2022 Course: B. Tech. Branch : CE Semester :IV Subject Code & Name: BTCVE404A (Numerical Methods in Engineering) Max Marks: 60 Date: Duration: 3 Hr.																				
	Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.																				
						(Level/CO)	Marks														
Q. 1	Solve Any Two of the following.																				
A)	Apply the Gauss-Jordan method to solve the equations. $x + y + z = 9$ $2x - 3y + 4z = 13$ $3x + 4y + 5z = 40$					CO1															
B)	Apply the Gauss-Seidel method to solve the equations. $20x + y - 2z = 17$ $3x + 20y - z = -18$ $2x - 3y + 20z = 25$					CO1															
C)	Solve the equations by using Cholesky method' $x + 2y + 3z = 5$ $2x + 8y + 22z = 6$ $3x + 22y + 82z = -10$					CO1															
Q.2	Solve Any Two of the following.																				
A)	Find the root of the equation $\cos x = xe^x$ using the bisection method correct to four decimal places.					CO2															
B)	Find a root of the equation $x^3 - 2x - 5 = 0$ using the secant method correct to three decimal places.					CO2															
C)	Using modified Euler's method, find an approximate value of y when $x = 0.3$, given that $dy/dx = x + y$ and $y = 1$ when $x = 0$.					CO2															
Q. 3	Solve Any One of the following.																				
A)	Using Newton's divided differences formula, evaluate $f(8)$ and $f(15)$ given: <table><tr><td>x</td><td>4</td><td>5</td><td>7</td><td>10</td><td>11</td><td>13</td></tr><tr><td>Y</td><td>48</td><td>100</td><td>294</td><td>900</td><td>1210</td><td>2028</td></tr></table>					x	4	5	7	10	11	13	Y	48	100	294	900	1210	2028	CO3	
x	4	5	7	10	11	13															
Y	48	100	294	900	1210	2028															
B)	Find the polynomial $f(x)$ by using Lagrange's formula and hence find $f(3)$ for <table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>5</td></tr><tr><td>y</td><td>2</td><td>3</td><td>12</td><td>147</td></tr></table>					x	0	1	2	5	y	2	3	12	147	CO3					
x	0	1	2	5																	
y	2	3	12	147																	

C)	Evaluate the integral $\int_0^1 \frac{x^2}{1+x^3} dx$ using Simpson's 1/3 rule. Compare the error with the exact value.	CO2													
Q.4	Solve Any Two of the following.														
A)	Calculate Mean and Standard Deviation for the data: <table><tr><td>x</td><td>0-10</td><td>10-20</td><td>20-30</td><td>30-40</td><td>40-50</td></tr><tr><td>$f(x)$</td><td>5</td><td>7</td><td>10</td><td>16</td><td>11</td></tr></table>	x	0-10	10-20	20-30	30-40	40-50	$f(x)$	5	7	10	16	11	CO3	
x	0-10	10-20	20-30	30-40	40-50										
$f(x)$	5	7	10	16	11										
B)	If P is the pull required to lift a load W by means of a pulley block, find a linear law of the form $P = mW + c$ connecting P and W , using the following data: <table><tr><td>P</td><td>12</td><td>15</td><td>21</td><td>25</td></tr><tr><td>W</td><td>50</td><td>70</td><td>100</td><td>120</td></tr></table>	P	12	15	21	25	W	50	70	100	120	CO3			
P	12	15	21	25											
W	50	70	100	120											
C)	Fit a second degree parabola to the following data: <table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>Y</td><td>1</td><td>1.8</td><td>1.3</td><td>2</td><td>6.3</td></tr></table>	x	0	1	2	3	4	Y	1	1.8	1.3	2	6.3	CO3	
x	0	1	2	3	4										
Y	1	1.8	1.3	2	6.3										
Q. 5	Solve Any One of the following.														
A)	Write the algorithm for Newton Raphson Method.	CO4													
B)	Write the algorithm for Euler's Method.	CO4													
C)	Write the algorithm for Trapezoidal Rule.	CO4													
	*** End ***														

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